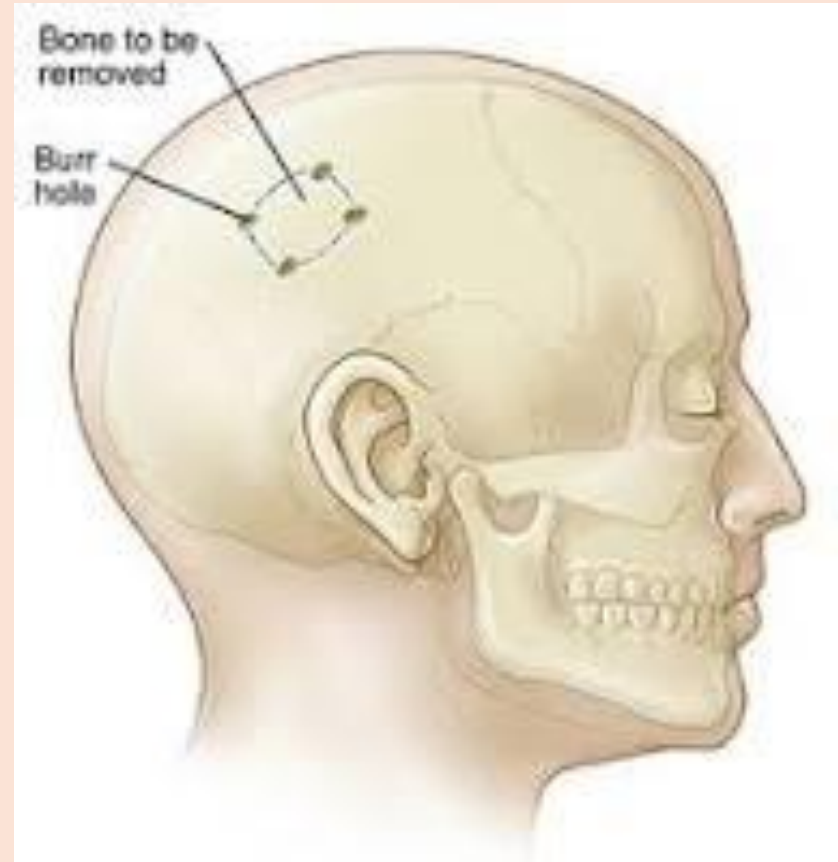


Craniotomy

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Outlines

- Definition
- Indications of Craniotomy
- Preoperative Care
- Postoperative Management of Craniotomy
- Nursing strategies to prevent postoperative complications
- Nursing diagnosis for patients post craniotomy

Pre test

Mention
indication of
craniotomy

Mention
complication
post craniotomy

Definition

- Is a surgical procedure where a neurosurgeon temporarily removes a piece of the skull (bone flap) to access the brain, treating conditions such as tumors, blood clots, aneurysms, or brain swelling
- Craniotomy is performed to remove a space-occupying abnormality such as a tumor, cyst, or vascular malformation.

Indications of Craniotomy

Common procedures include:

tumor resection or removal,

cerebral decompression,

evacuation of hematoma or abscess,

clipping or removal of an aneurysm

or Arteriovenous Malformation (AVM).

Preoperative Care:



Protection of the integrity of the CNS is a major priority of care for the patient awaiting craniotomy;



- Optimal arterial oxygenation,



hemodynamic stability,



and cerebral perfusion are essential for maintaining adequate cerebral oxygenation.



Management of seizure activity is essential for controlling metabolic needs.

Evaluation of endocrine function is necessary to prevent major intra-operative and postoperative complications.

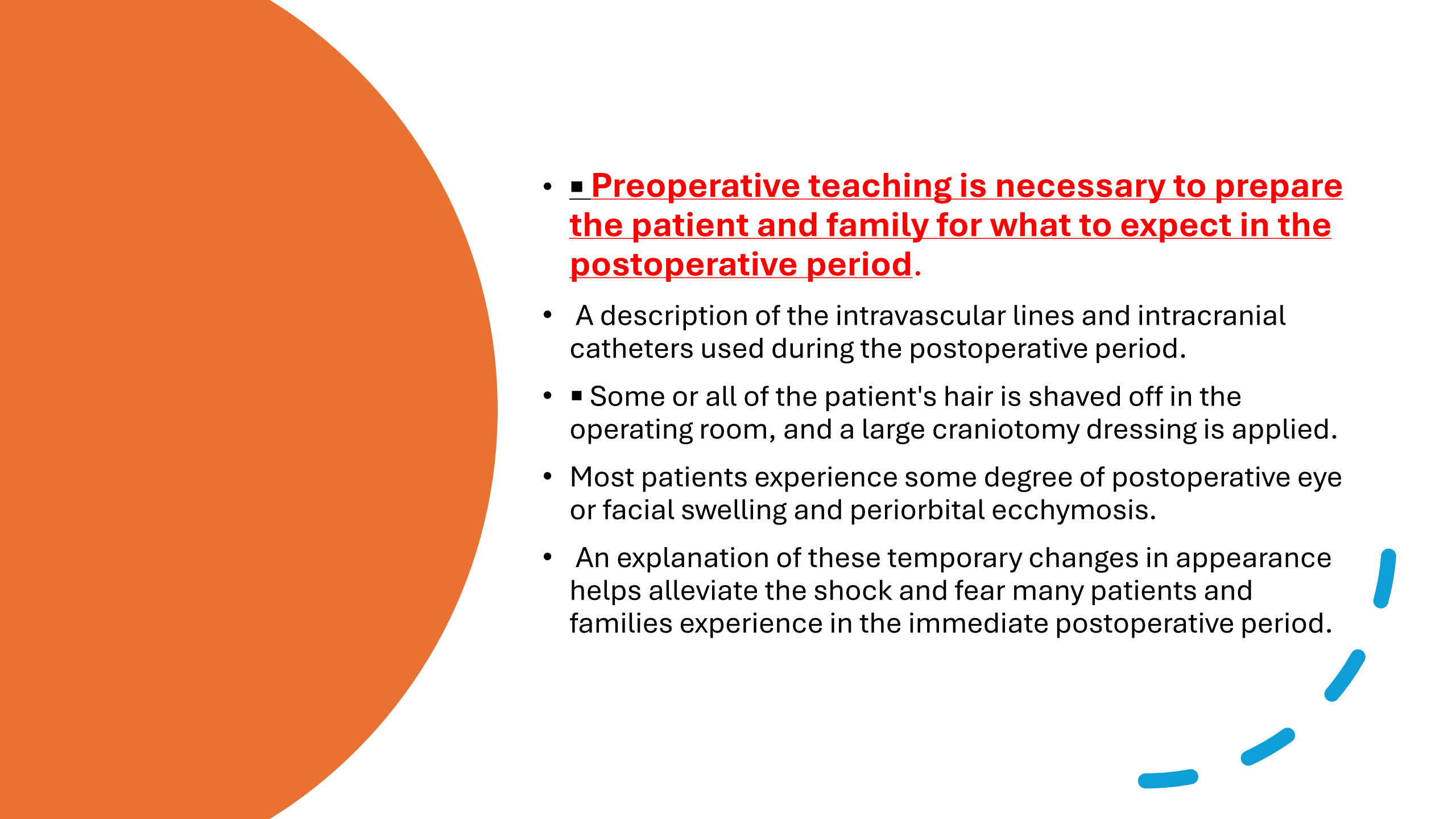
▪ Use of routine preoperative studies may include a Complete Blood Count (CBC);

tests for Blood Urea Nitrogen (BUN), creatinine,

and Fasting Blood Glucose (FBG).

a chest radiograph; and an electrocardiogram.

Blood type and cross match may also be ordered.

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- **■ Preoperative teaching is necessary to prepare the patient and family for what to expect in the postoperative period.**
 - A description of the intravascular lines and intracranial catheters used during the postoperative period.
 - ■ Some or all of the patient's hair is shaved off in the operating room, and a large craniotomy dressing is applied.
 - Most patients experience some degree of postoperative eye or facial swelling and periorbital ecchymosis.
 - An explanation of these temporary changes in appearance helps alleviate the shock and fear many patients and families experience in the immediate postoperative period.

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- All patients undergoing craniotomy require instruction to avoid activities known to provoke sudden changes in ICP.
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These activities include bending, lifting, straining, and the Valsalva maneuver.

Teaching the patient to continue to breathe deeply through the mouth during all position changes is an effective measure to prevent increase ICP.

The patient undergoing surgery requires preparation for the sensations associated with nasal packing.

The patient often awakens with alarm because of the inability to breathe through the nose.

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- Preoperative instruction in mouth breathing and avoidance of coughing, sneezing, and blowing of the nose facilitates postoperative cooperation.
 - The psychosocial issues associated with neurosurgery cannot be over emphasized.
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For some patients may be fear from permanent neurological impairment or fear of death.

The patient and family must be given the opportunity to express their fears and concerns jointly.

Postoperative Management of Craniotomy Management directed toward the prevention of complications including the following:

Intracranial Hypertension

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- **Postoperative cerebral edema** is expected to peak 48 to 72 hours after surgery.
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If the bone flap is not replaced at the time of surgery, intracranial hypertension produces bulging at the surgical site.

Close monitoring of the surgical site is important so that integrity of the incision can be maintained.

- Post craniotomy management of intracranial hypertension is usually accomplished through CSF drainage, patient positioning, hyperventilation and steroid administration.

Surgical Hemorrhage Surgical hemorrhage after a transcranial procedure is manifested by signs and symptoms of increasing ICP.

Hemorrhage post craniotomy may be evident from external drainage, the patient's complaint of persistent postnasal drip, or excessive swallowing.

Loss of vision after pituitary surgery indicates an evolving hemorrhage. Postoperative hemorrhage requires surgical re-exploration.

Fluid Imbalance

▪ **Fluid imbalance** in the post craniotomy patient usually results from a disturbance in production or secretion of Antidiuretic Hormone (ADH).

ADH is secreted by the posterior pituitary gland.

Intra-operative manipulation of the pituitary gland or hypothalamus or postoperative edema can result in insufficient ADH.

This condition is known as Diabetes Insipidus (DI).

- The polyuria associated with DI is often more than 200 mL/hour. Urine specific gravity of 1.005 or less and elevated serum osmolality provide evidence of insufficient ADH
 - DI is usually self-limiting, and fluid replacement is the only required therapy.
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It may be necessary to administer vasopressin intravenously to control the loss of fluid.

Cerebrospinal Fluid Leak

- **Leakage of Cerebrospinal** Fluid (CSF) results from an opening in the subarachnoid space, as evident by clear fluid draining from the surgical site.

- To differentiate CSF drainage from postoperative serous drainage, a specimen is tested for glucose content.

A CSF leak is confirmed by glucose values of 30 mg/dL or greater.

- Management of the patient with a CSF leak includes bed rest and head elevation.

Lumbar puncture or placement of a lumbar subarachnoid catheter may be used to reduce CSF pressure until the dura heals.

- **The risk of meningitis associated with CSF leak** often necessitates surgical repair to reseal the opening.

Deep Vein Thrombosis

- Risk factors significantly associated with neurosurgery, including preoperative leg weakness, longer preoperative ICU stay, longer recovery room time, longer postoperative ICU stay, more days of bed rest, and delay of postoperative mobility and activity.
- Clinical manifestations of DVT include leg or calf pain, edema, localized tenderness, and pain with dorsiflexion or plantar flexion (Homans' Sign).
- The primary treatment for DVT is prophylaxis by intermittent pneumatic compression boots or stockings are effective in reducing the incidence of DVT.

Low-dose unfractionated heparin or low-molecular-weight heparin may also be used prophylactically in high risk patients.

Postoperative Nursing Management Preserving Adequate Cerebral Perfusion Nursing interventions to preserve cerebral perfusion include:

patient positioning, fluid management, and avoidance of postoperative vomiting and fever.

Positioning: The head of the bed should be elevated 30 to 45 degrees at all times to reduce the incidence of hemorrhage, facilitate venous drainage, and control ICP.

- Other positioning measures to control ICP include maintaining the patient's head in a neutral position at all times and avoiding neck and hip flexion .

Most craniotomy patients can be turned from side to side within these restrictions, using pillows for support, except in some cases of extensive tumor removal, cranioplasty, and when the bone flap is not replaced.

Fluid Management:

- Hourly monitoring of fluid intake and output facilitates early identification of fluid imbalance.

- Urine specific gravity must be measured if DI is suspected.

- Fluid restriction may be ordered as a routine measure to lessen the severity of cerebral edema or as treatment for the fluid and electrolyte imbalances associated with SIADH.

Vomiting and Fever:

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- Postoperative vomiting must be avoided to prevent increase in ICP and possibly surgical hemorrhage.
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- Antiemetics are administered as soon as nausea is apparent.
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- Early resumption of nutrition in the neurosurgical patient is beneficial.
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If the patient is unable to eat, enteral feeding delivered through a feeding tube is the preferred method of nutritional support and can be initiated as early as 24 hours after surgery.

Postoperative fever:

Fever may adversely affect ICP and increase the metabolic needs of the brain.

Acetaminophen is administered orally, rectally, or through a feeding tube.

External cooling measures, such as a hypothermia blanket, may be necessary.

Promoting Arterial Oxygenation Routine pulmonary care is used to maintain airway clearance and prevent pulmonary complications.

To prevent dangerous elevations in ICP, this care must be performed with the use of proper technique and at time intervals that are adequately spaced from other patient care activities.

If pulmonary complications do arise, consideration must be given to maintaining adequate oxygenation during repositioning.

Providing Comfort and Emotional Support Pain management:

in the post-craniotomy patient primarily involves control of headache.

Small doses of intravenous morphine are used in the critical care setting.

As soon as oral analgesics can be tolerated, acetaminophen with codeine is used.

Both analgesics cause constipation.

Administration of stool softeners and initiation of a bowel program are important components of post-craniotomy care.

Constipation is hazardous, because straining to have a bowel movement can create significant elevations in blood pressure and ICP.

Maintaining Surveillance for Complications:

The postoperative neurosurgical patient is at risk for infection, corneal abrasions, and injury from falls or seizures.

Infection: Care of the incision and surgical dressings change as a physician's order.

Often, a drain is left in place to facilitate decompression of the surgical site.

All drainage devices must be secured to the dressing to prevent displacement with patient movement.

Sterile technique is required to prevent infection and resultant meningitis.



- **Corneal Abrasions:**

- Routine eye care may be necessary to prevent corneal drying and ulceration.
- Periorbital edema interferes with normal blinking and eyelid closure, which are essential to adequate corneal lubrication.
- Saline drops are instilled every 2 hours.
- If the patient remains in a coma state, covering the eyes with a polyethylene film extending over the orbits and eyebrows may be beneficial.

- **Injury:**

- The post-craniotomy patient may experience periods of altered mentation.
- Protection from injury may require the use of restraint devices. The side rails of the bed must be padded to protect the patient from injury.
- Having a family member stay at the bedside or use of music therapy is often helpful to keep the patient calm during periods of restlessness.
- In rare circumstances, neuromuscular blockade and sedation may be necessary to control patient activity and metabolic needs on a short term basis.

• Initiating Early Rehabilitation

- Increased activity, including ambulation, is begun as soon in the postoperative period as tolerated by the patient.
- Transfer to a general care or rehabilitation unit is usually accomplished as soon as the patient is considered to be stable and free of complications.
- Educating Patient and Family As the patient moves toward discharge, teaching focuses on medication instructions; incisional care, including the signs of infection; and the signs and symptoms of increased ICP.

Examples of NURSING DIAGNOSES Post Craniotomy

- Decreased Intracranial Adaptive Capacity related to failure of normal intra-cranial compensatory mechanisms.

- Ineffective Cerebral Tissue Perfusion.

- Acute Pain related to transmission and perception of cutaneous, visceral, muscular, or ischemic impulses.

- Disturbed Body Image related to actual change in body structure, function, or appearance.

PATIENT EDUCATION PLAN Post Craniotomy

Before the patient's discharge, the patient should be able to teach back the following topics:

Before Surgery

- Pathophysiology and expected outcome of underlying disease

- Need for intensive care management after surgery

- Routine preoperative surgical care After Surgery

- Routine postoperative surgical care

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Discharge medications purpose, dosage, and side effects

- Incisional care

- Signs and symptoms of infection

- Signs and symptoms of increased intracranial pressure

- Measures to compensate for residual deficits

- Basic rehabilitation techniques

- Importance of participating in neurologic rehabilitation program

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Post test

Mention complication
of craniotomy

Mention post operative
care to patients post
craniotomy



Any question



Thank you